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The Political Economy of a Coal Phase-Out

A war of attrition over three decades — and counting

Diego Pardow
Universidad de Chile

The Chilean coal phase-out: a success story

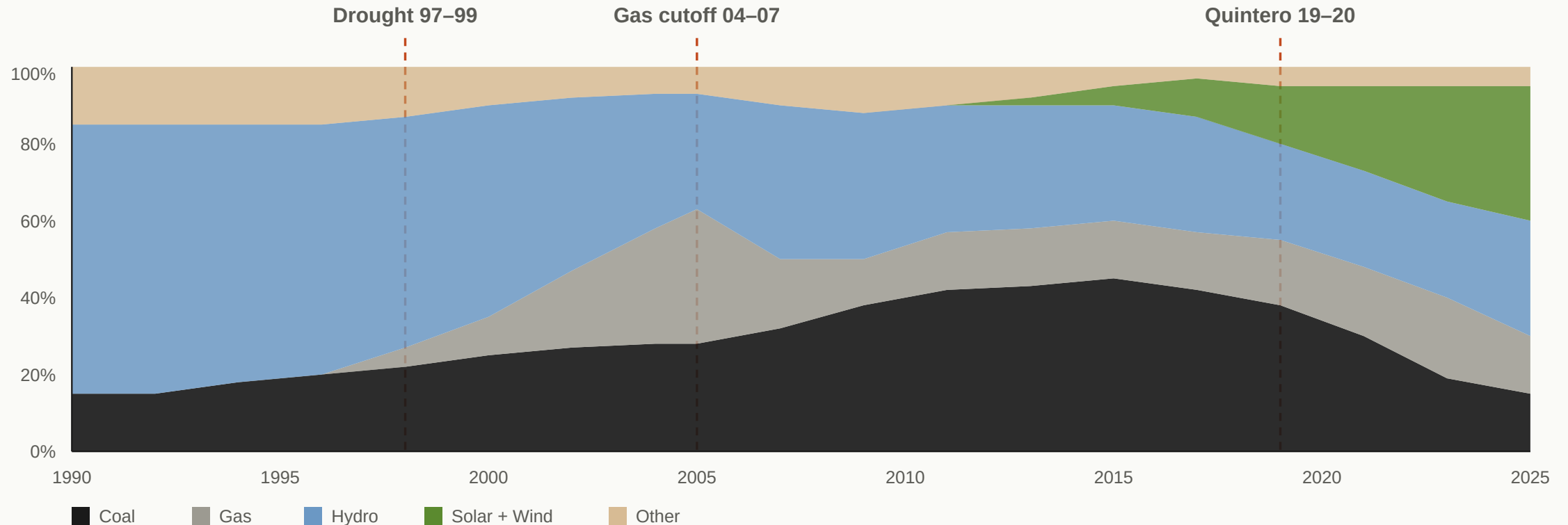
In 2019, the Chilean government and the four main coal generators signed a Voluntary Phase-Out Agreement: all 28 coal-fired units committed to retirement by 2040, oldest plants first. By 2025, 11 units had already been retired and 9 more were scheduled to close or convert by end of 2026 — 70% of the 2019 coal capacity gone in less than seven years.

Year	Solar + Wind	Coal	Gas	Hydro	Other	Emissions	Price
2010	<1%	40%	15%	35%	10%	0.55	150
2012	<1%	45%	15%	30%	10%	0.55	160
2014	5%	45%	15%	30%	5%	0.50	150
2016	10%	40%	20%	25%	5%	0.40	125
2018	15%	40%	15%	25%	5%	0.35	110
2020	20%	30%	20%	25%	5%	0.30	100
2022	30%	20%	20%	25%	5%	0.25	110
2024	35%	15%	15%	30%	5%	0.20	110

Power-sector emission factor fell by 63% in a decade. Some of the lowest wind and solar auction prices in the world. No subsidy spikes, no supply disruptions, no major tariff shocks. Chile is now routinely cited at COP26, COP28, and IEA reports as a Global South success case.

Each row sums to 100%. "Other" = biomass, oil-derivates, geothermal, and (recently) storage. Emissions in tCO₂/MWh. Price in USD/MWh, constant 2024.

Three traumas that shaped Chilean energy policy



1997-99 — Drought

Severe drought collapses hydro generation. The first democratic governments face energy as a brake on the economy. The state responds with emergency thermal capacity. Coal and gas enter together as firm baseload. **The trauma legitimizes fossil expansion.**

2004-07 — Gas cutoff

Argentina cuts gas exports without notice. The state underwrites LNG terminals (operational 2009-11, a precedent for Europe's response to Russian gas). In the five-year interim, coal — not gas — gains market share. **The trauma legitimizes coal as "secure" baseload.**

2018-20 — Quintero crisis

School poisoning. Coal is no longer the solution — it has become the problem. The regulatory response is forced by public opinion and Supreme Court rulings. **Renewables, not gas, take coal's market share.**

Regulation is historical, not just optimal. Each trauma reopened a narrative channel — and someone learned to capture it.

Why capture is structural, not episodic

Coal plants are highly specific assets (Williamson 1985). Their value collapses outside coal-burning.

This specificity creates a quasi-rent shared between capital (the generator) and labor (the unionized workforce concentrated in sacrifice zones — Mejillones, Tocopilla, Coronel, Huasco, Puchuncaví — and the local supply chains that depend on them).

Capital and labor lobby together — not from shared ideology, but from a common rent to defend. **This is the Ackerman-Hassler coalition (1981)**, originally identified in the U.S. Clean Air Act of 1977.

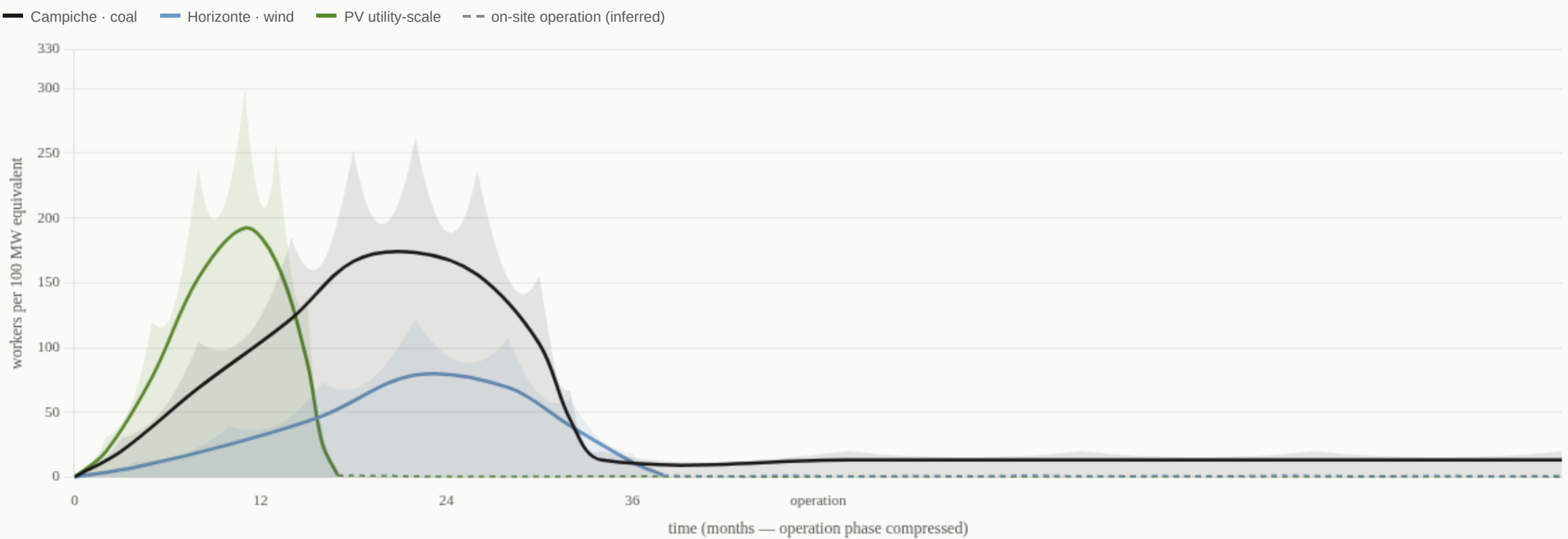
Renewables are cheaper and create **mostly temporary construction jobs** — high peak, fast decline, leaving local economies with whiplash rather than dependence. But coal sustains **more permanent, unionized, geographically concentrated employment** — a structural alteration that the local economy adapts to, and depends on. And that is what holds the coalition together.

Each new regulatory instrument creates a new lobbying surface. The coalition adapts. The instrument absorbs the capture. The plant keeps running.

	Construction	O&M
Coal / Gas	Multi-year on-site build, workforce settles in the community	Permanent on-site workforce, unionized
Solar / Wind	12–18 month on-site build, mostly contractor crews	Minimal on-site presence; cleaning and inspection subcontracted

Comparison for utility-scale projects of comparable installed capacity. Construction = on-site build phase (excludes off-site manufacturing of components). O&M = Operation and Maintenance over the asset life.

The labor footprint that built the coalition



Coal mobilizes 4x more total labor than wind & solar, with half of the jobs being permanent. Ninety percent of renewable jobs vanish once construction ends.

Three rounds of capture

2000s — Building the rent. Following the gas crisis, the capital-labor coalition lobbies for a new generation fleet — modern coal units in Mejillones, Huasco, and Puchuncaví, framed as the response to the security-of-supply trauma. The state authorizes 28 units, many of them built after 2005. Capital is invested at scale; unionized workforces and local supply chains consolidate around each port complex. The quasi-rent is created. By 2014, coal accounts for 45% of generation.

2010s — Protecting the rent. The Quintero school poisoning forces a regulatory response. The capital-labor coalition does not block it — it shapes it. A green tax is introduced (US\$5/ton CO₂), but with a compensation mechanism that passes the cost to renewable generators through the spot market — the very competitors the tax was supposed to favor. The corrective signal is reversed; the plant keeps burning coal, and renewables subsidize it. The compensation is later litigated and struck down — but only after years of operating under the captured version.

2020s — Exit strategies. Renewables now displace coal on cost. Capital can no longer protect the rent — it must liquidate it. Four exit strategies emerge: (i) the Strategic Reserve State (ERE), keeping "closed" plants on paid standby; (ii) sale to specialized funds without global brands to protect; (iii) conversion to gas or green ammonia (e.g., IEM Mejillones), which preserves the asset and changes only the fuel; (iv) conversion to BESS, which genuinely transforms the business model. The first three preserve the rent in different forms; only the fourth liquidates it. The instrument absorbs the difference.

The instruments change. The coalition adapts. Each round distributes rents to capital and labor in a different way — but always preserves the asset, and always delays the closure.

Future challenges

The capital-labor coalition never gives up, never surrenders

Political reframing under the new government. In February 2025 Chile suffered its worst blackout in 15 years, followed in April by the Iberian blackout — both caused by transmission and voltage-control issues, not by renewables. In both cases, the public debate was reframed: closures had been "too fast," firm baseload was being lost, sovereignty was at risk. Cost, sovereignty, and supply reliability are now being used to renegotiate rounds that were considered settled.

Unfinished business in the closed plants. Sites retired since 2019 still lack environmental remediation. The social transition cost (Plan Coronel, Plan Tocopilla) has been transferred to the state, not absorbed by the firms.

Gas, the next coal. The same playbook is being applied — gas as "transition fuel," LNG terminals seeking long-term contracts, stranded-asset risk migrating from coal to gas.

Closure timeline at risk. Plants once scheduled to retire are being renegotiated under "energy security" framing. The ERE, sold as transitional, is becoming permanent infrastructure.

The slope of the transition becomes steeper

Remaining hydrocarbons get more expensive. Coal and LNG prices have settled at roughly 1.5–2× pre-COVID levels post-Ukraine. Even as the fossil share falls, what remains costs more — eroding the price promise of the transition.

Macro instability and inflation. Prices in Chilean pesos have risen despite the renewable build-out. The benefit measured in real local terms is much smaller than the headline LCOE decline suggests.

New sacrifice zones. Solar and wind use far more land than coal. We are moving from bay-scale sacrifice zones (Mejillones, Coronel) to region-scale ones — Atacama covered in panels, Magallanes in turbines, transmission corridors crossing them all.

What Chile actually teaches

1

Transitions are not designed. They are fought.

The Chilean success was not technocratic excellence. It was political endurance under permanent distributional conflict — sustained across three decades, four governments, and shifting coalitions on every side.

2

Capture is structural, not episodic.

Each instrument grows its own coalition. The 2000s built the rent, the 2010s protected it, the 2020s liquidate it through four different exit strategies. Designing the next instrument without modeling this dynamic is naive.

3

Victories are reversible.

The day the agreement is signed is the day the front line moves. The 2019 phase-out agreement did not end the conflict — it relocated it. Each closure produces the next stranded asset, the next political reframing, the next round.

Learn from Chile — but learn the right thing.

Learn how to prepare policies for a fight, not just desk-design.

Further reading

- **Ackerman, B., & Hassler, W. T. (1981).** *Clean Coal/Dirty Air: Or How the Clean Air Act Became a Multibillion-Dollar Bail-Out for High-Sulfur Coal Producers*. Yale University Press.
- **Supreme Court of Chile (2019).** *Quintero-Puchuncaví, Rol 5888-2019*.
observatoriop10.cepal.org/es/jurisprudencia/sentencia-la-corte-suprema-chile-rol-no-5888-2019
- **Feng, K., Song, K., Viteri, A., Liu, Y., & Vogt-Schilb, A. (2023).** *National and Local Labor Impacts of Coal Phase-Out Scenarios in Chile*. *Journal of Cleaner Production*, 414, 137399.
- **Castillo, P., Aguad, M., Lorca, Á., Cordova, S., & Negrete-Pincetic, M. (2024).** *Coal Phase-Out and Carbon Tax Analysis with Long-Term Planning Models: A Case Study for the Chilean Electric Power System*. *Energies*, 17(21), 5263.
- **Pardow, D. (forthcoming).** *Regulación de Mercados: Casos y Materiales*, Ch. 2. Universidad de Chile.